

Student Number: _____ Teacher Name: _____

PENRITH SELECTIVE HIGH SCHOOL
MATHEMATICS DEPARTMENT



2026 TERM 3 TRIAL HSC EXAMINATION

YEAR 12 Mathematics Extension 1

General Instructions:

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided with this examination paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- No correction tape or white out allowed.

Total marks: 70

Section I – 10 marks (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 5–8)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Question Topic	Multiple Choice	Q.11	Q.12	Q.13	Q.14	Total
Combinatorics	7 / 1		a / 2			/ 3
Functions	5 / 1	a / 2		a / 3		/ 6
Proof				b / 4		/ 4
Trigonometric Functions	1, 6, 9 / 3		b, c / 5+2			/ 10
Calculus	2, 3 / 2	b, c, d / 2+4+3	d, e / 3+3	c / 4		/ 21
Vectors	8, 10 / 2			d / 4	a, b / 5+10	/ 21
Binomial Distribution	4 / 1	e / 4				/ 5
Total	/ 10	/ 15	/ 15	/ 15	/ 15	/ 70

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Find the domain and range of $y = \frac{1}{2} \cos^{-1}(3 - x)$.

- A. The domain is $0 \leq x \leq \frac{\pi}{2}$, and the range is $2 \leq y \leq 4$.
- B. The domain is $-4 \leq x \leq -2$, and the range is $0 \leq y \leq \frac{\pi}{2}$.
- C. The domain is $2 \leq x \leq 4$, and the range is $0 \leq y \leq \frac{\pi}{2}$.
- D. The domain is $2 \leq x \leq 4$, and the range is $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$.

2 Find $\int \frac{1}{\sqrt{9 - 16x^2}} dx$.

- A. $\frac{1}{4} \sin^{-1} \frac{16x^2}{9} + C$
- B. $\frac{1}{4} \cos^{-1} \frac{16x^2}{9} + C$
- C. $\frac{1}{4} \sin^{-1} \frac{3}{4x} + C$
- D. $-\frac{1}{4} \cos^{-1} \frac{4x}{3} + C$

3 Which function is a general solution of the differential equation $y'' + 4y = 0$?

- A. $y = Ae^{2x} + Be^{-2x}$
- B. $y = A \sin x + B \cos x$
- C. $y = A \sin 2x + B \cos 2x$
- D. $y = A \sin 4x + B \cos 4x$

- 4 A company manufactures microchips. Typically, about 3% of all microchips produced are defective. In a shipment of 80 microchips, find the probability that at least three microchips are defective.
- A. 0.5681
B. 0.9098
C. 0.4319
D. 0.5193
- 5 The remainder when the polynomial $P(x)$ is divided by $(x + 2)(x - 5)$ is $3x + 1$. What is the remainder when $P(x)$ is divided by $x + 2$?
- A. 1
B. -5
C. -9
D. 16
- 6 Let $t = \tan \frac{x}{2}$. What values of t satisfy the equation $2 \cos x + \sin x = -2$?
- A. $t = 2$
B. $t = -2$ and $t = \pi$
C. $t = 2$ and $t = \pi$
D. $t = -2$
- 7 Let $P(x) = 1 - x^2 + x^4 - x^6 + x^8 - x^{10}$. What is the coefficient of x^7 in the expansion of $P(x - 1)$?
- A. 0
B. 128
C. 112
D. -128

- 8 What is the unit vector in the direction of $\overrightarrow{AB}$, where A and B are the points $(-2, 1)$ and $(6, 3)$, respectively?
- A. $\frac{4}{\sqrt{17}}\mathbf{i} + \frac{1}{\sqrt{17}}\mathbf{j}$
- B. $8\mathbf{i} + 2\mathbf{j}$
- C. $\frac{2}{17}\mathbf{i} + \frac{1}{34}\mathbf{j}$
- D. $\frac{2}{\sqrt{5}}\mathbf{i} + \frac{1}{\sqrt{5}}\mathbf{j}$
- 9 Given that $\tan x = k$, where $\pi < x < \frac{3\pi}{2}$, determine the value of $\sin \frac{x}{2}$.
- A. $-\sqrt{\frac{1}{2}\left(1 + \frac{k}{\sqrt{1+k^2}}\right)}$
- B. $\sqrt{\frac{1}{2}\left(1 + \frac{k}{\sqrt{1+k^2}}\right)}$
- C. $-\sqrt{\frac{1}{2}\left(1 + \frac{1}{\sqrt{1+k^2}}\right)}$
- D. $\sqrt{\frac{1}{2}\left(1 + \frac{1}{\sqrt{1+k^2}}\right)}$
- 10 If $|\underline{a} - \underline{b}| = |\underline{a}| + |\underline{b}|$, where $\underline{a} \neq \underline{0}$ and $\underline{b} \neq \underline{0}$, which one of the following must be true?
- A. $|\underline{a}| = |\underline{b}|$
- B. $\underline{a}$ is parallel to $\underline{b}$
- C. $\underline{a} = \underline{b}$
- D. $\underline{a} \perp \underline{b}$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section.

- Answer the questions in the spaces provided.
 - Your responses should include relevant mathematical reasoning and/or calculations.
-

Question 11 (15 marks) Use a separate Writing Booklet

- (a) Solve $|4x - 3| > 11$. 2
- (b) A cup of coffee is heated to 90°C . The room temperature is maintained at a constant 20°C , and the coffee cools to 50°C after 8 minutes. The temperature of the coffee can be modelled by the equation $T = 20 + 70e^{-kt}$, where T is the temperature of the coffee in degrees Celsius, t is the time in minutes since the coffee was placed in the room, and k is a constant. 2
- Determine the time taken for the temperature of the coffee to reach 25°C .
Give your answer correct to 1 decimal place.
- (c) A solid is formed by rotating the region bounded by the curves $y = 4 - x^2$ and $y = 2 - x$ about the x -axis. Find the exact value of the volume of the solid. 4
- (d) (i) Complete the square for $x^2 - 6x + 13$, expressing the answer in the form $[f(x)]^2 + a$, where a is a constant and $f(x)$ is a linear function of x . 1
- (ii) Hence, find $\int \frac{1}{x^2 - 6x + 13} dx$. 2
- (e) A standard pack of 52 playing cards consists of 13 hearts, 13 diamonds, 13 clubs and 13 spades. Fifteen cards are drawn with replacement from the pack.
- (i) Let X be the number of diamonds drawn. Find the mean and standard deviation of X , giving your answers correct to 2 decimal places, where applicable. 2
- (ii) Determine the probability that X is within one standard deviation of the mean. 2
Give your answer correct to 3 decimal places.

Question 12 (15 marks) Use a separate Writing Booklet

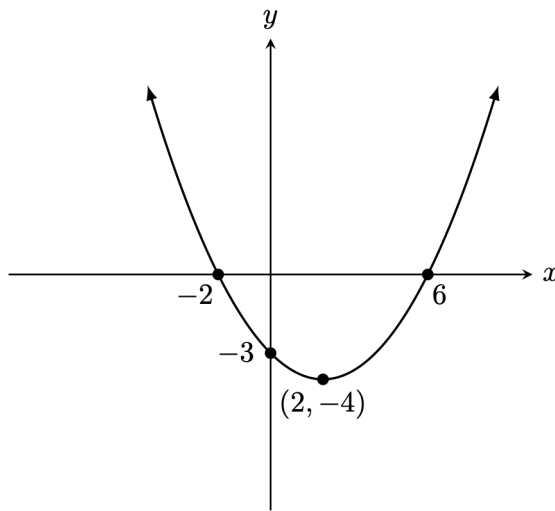
- (a) There are 5 men and 6 women to choose from. In how many ways can 3 men and 5 women be selected and arranged in a row? **2**
- (b) Consider the function $f(x) = \sqrt{3} \cos x + 3 \sin x$ for $0 \leq x \leq 2\pi$.
- (i) By first expanding $\cos(x - \alpha)$, express $f(x)$ in the form $R \cos(x - \alpha)$. **3**
- (ii) Hence, solve $\sqrt{3} \cos x + 3 \sin x = 1$, correct to 2 decimal places. **2**
- (c) Express $\sin\left(2 \cos^{-1} \frac{3}{4}\right)$ in the form $a\sqrt{b}$, where a and b are rational numbers. **2**
- (d) A drone is flying horizontally at a constant height of 120 m above the ground. **3**
It moves horizontally away from a person at a constant speed of 6 m/s.
Determine the rate at which the distance between the drone and the person is increasing when the drone is 150 m from the person.
- (e) Using the substitution $u = \sqrt{6x - 1}$, show that **3**
$$\int \frac{x}{\sqrt{6x - 1}} dx = \frac{1}{27} (3x + 1) \sqrt{6x - 1} + C.$$

Question 13 (15 marks) Use a separate Writing Booklet

(a) The graph of $y = f(x)$ is shown below.

3

Sketch the graph of $y = |f(|x|)|$, clearly showing all key features.



(b) Prove by mathematical induction that $4(3^n) + 8^n$ is divisible by 5 for all integers $n \geq 1$.

4

(c) Consider the differential equation $\frac{dy}{dx} = \frac{y(1-x)}{x}$.

(i) State any constant solutions.

1

(ii) Find the general solution of the differential equation by separating the variables.

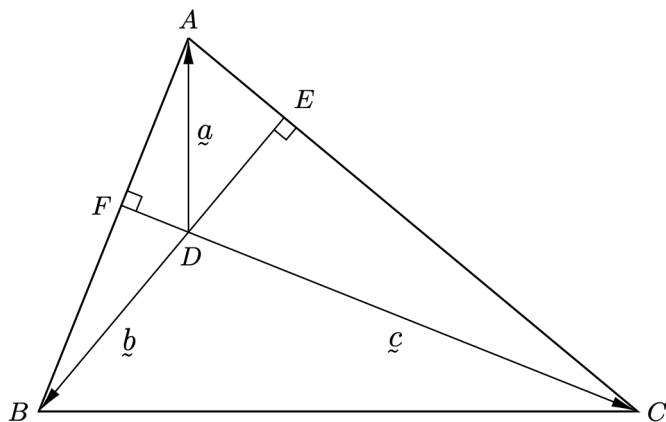
3

(d) Find all possible values of k given that the length of the projection of $\mathbf{u}$ onto $\mathbf{v}$ is $\frac{108}{17}$,
where $\mathbf{u} = k\mathbf{i} + 9\mathbf{j}$ and $\mathbf{v} = 15\mathbf{i} - 8\mathbf{j}$.

4

Question 14 (15 marks) Use a separate Writing Booklet

- (a) The $\triangle ABC$ shown below has altitudes BE and CF , where an altitude is a line segment drawn from a vertex perpendicular to the opposite side. The altitudes intersect at D . Let $\overrightarrow{DA} = \underline{a}$, $\overrightarrow{DB} = \underline{b}$ and $\overrightarrow{DC} = \underline{c}$.



- (i) Show that $\underline{a} \cdot \underline{b} = \underline{a} \cdot \underline{c} = \underline{b} \cdot \underline{c}$. 3
- (ii) Hence, or otherwise, prove that the three altitudes of $\triangle ABC$ are concurrent; that is, they all pass through a single point. 2
- (b) A ball is projected from the base of a cliff with initial speed u m/s at an angle of θ to the horizontal. At the same time, a second ball is projected horizontally with initial speed w m/s from the top of the same cliff, which is h m high. Take the base of the cliff as the origin $(0, 0)$. The two balls collide.
- (i) Show that the position vector of the first ball is 3
 $\mathbf{r}_1(t) = (ut \cos \theta)\mathbf{i} + \left(ut \sin \theta - \frac{g}{2}t^2\right)\mathbf{j}$.
- (ii) It is given that $\theta = \tan^{-1} 3$ and the position vector of the second ball is 2
 $\mathbf{r}_2(t) = wt\mathbf{i} + \left(h - \frac{g}{2}t^2\right)\mathbf{j}$.
 Show that $\cos \theta = \frac{1}{\sqrt{10}}$, and hence deduce that $u = \sqrt{10}w$.
- (iii) Hence, prove that the balls collide at time $t = \frac{h}{3w}$, and that the height of 2
 the balls at the point of collision is $h - \frac{gh^2}{18w^2}$.
- (iv) At the instant of collision, the velocity vectors of the two balls are perpendicular. 3
 Show that $h = \frac{3(3 \pm \sqrt{5})w^2}{2g}$.

End of paper

Student Number: **Feedback & Solution** Teacher Name: _____

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Calculus	2, 3 / 2	b, c, d / 2+4+3	d, e / 3+3	c / 4		/ 21
Vectors	8, 10 / 2			d / 4	a, b / 5+10	/ 21
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Section I

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- B. The domain is $-4 \leq x \leq -2$, and the range is $0 \leq y \leq \frac{\pi}{2}$.
- C. The domain is $2 \leq x \leq 4$, and the range is $0 \leq y \leq \frac{\pi}{2}$.
- D. The domain is $2 \leq x \leq 4$, and the range is $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$.

Domain: $-1 \leq 3 - x \leq 1$

$$-4 \leq -x \leq -2$$

$$4 \geq x \geq 2$$

$$\therefore 2 \leq x \leq 4$$

Range: $0 \leq \cos^{-1}(3 - x) \leq \pi$

$$\therefore 0 \leq \frac{1}{2} \cos^{-1}(3 - x) \leq \frac{\pi}{2}$$

$$\therefore 0 \leq y \leq \frac{\pi}{2}$$

Answer: C

2 Find $\int \frac{1}{\sqrt{9 - 16x^2}} dx$.

A. $\frac{1}{4} \sin^{-1} \frac{16x^2}{9} + C$

B. $\frac{1}{4} \cos^{-1} \frac{16x^2}{9} + C$

C. $\frac{1}{4} \sin^{-1} \frac{3}{4x} + C$

D. $-\frac{1}{4} \cos^{-1} \frac{4x}{3} + C$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + C = -\cos^{-1} \frac{f(x)}{a} + C$$

$$\begin{aligned} \int \frac{1}{\sqrt{9 - 16x^2}} dx &= \int \frac{1}{\sqrt{(3)^2 - (4x)^2}} dx = \frac{1}{4} \int \frac{4}{\sqrt{(3)^2 - (4x)^2}} dx = -\frac{1}{4} \int \frac{-4}{\sqrt{(3)^2 - (4x)^2}} dx \\ &= -\frac{1}{4} \cos^{-1} \frac{4x}{3} + C \end{aligned}$$

Answer: D

3 Which function is a general solution of the differential equation $y'' + 4y = 0$?

- A. $y = Ae^{2x} + Be^{-2x}$
- B. $y = A \sin x + B \cos x$
- C. $y = A \sin 2x + B \cos 2x$
- D. $y = A \sin 4x + B \cos 4x$

For option C,

$$y = A \sin 2x + B \cos 2x$$

$$y' = 2A \cos 2x - 2B \sin 2x$$

$$y'' = -4A \sin 2x - 4B \cos 2x$$

$$y'' = -4(A \sin 2x + B \cos 2x)$$

$$y'' = -4y$$

$$\therefore y'' + 4y = 0$$

Answer: C

4 A company manufactures microchips. Typically, about 3% of all microchips produced are defective. In a shipment of 80 microchips, find the probability that at least three microchips are defective.

- A. 0.5681
- B. 0.9098
- C. 0.4319
- D. 0.5193

Let X = number of defective microchips.

$$P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

$$= 1 - [(0.97)^{80} + {}^{80}C_1(0.03)(0.97)^{79} + {}^{80}C_2(0.03)^2(0.97)^{78}]$$

$$= 0.4319$$

Answer: C

5 The remainder when the polynomial $P(x)$ is divided by $(x + 2)(x - 5)$ is $3x + 1$. What is the remainder when $P(x)$ is divided by $x + 2$?

- A. 1
- B. -5
- C. -9
- D. 16

$$P(x) = (x + 2)(x - 5) Q(x) + (3x + 1)$$

$$\begin{aligned} P(-2) &= (-2 + 2)(-2 - 5) Q(-2) + (3(-2) + 1) \\ &= -5 \end{aligned}$$

Answer: B

6 Let $t = \tan \frac{x}{2}$. What values of t satisfy the equation $2 \cos x + \sin x = -2$?

- A. $t = 2$
- B. $t = -2$ and $t = \pi$
- C. $t = 2$ and $t = \pi$
- D. $t = -2$

$$2 \cos x + \sin x = -2$$

$$\frac{2(1-t^2)}{1+t^2} + \frac{2t}{1+t^2} = -2$$

$$2(1-t^2) + 2t = -2(1+t^2)$$

$$2 - 2t^2 + 2t = -2 - 2t^2$$

$$2t = -4$$

$$t = -2$$

Answer: D

7 Let $P(x) = 1 - x^2 + x^4 - x^6 + x^8 - x^{10}$.
What is the coefficient of x^7 in the expansion of $P(x-1)$?

- A. 0
- B. 128
- C. 112
- D. -128

$$P(x-1) = 1 - (x-1)^2 + (x-1)^4 - (x-1)^6 + (x-1)^8 - (x-1)^{10}$$

$$(x-1)^8 = \sum_{r=0}^8 {}^8C_r x^{8-r} (-1)^r$$

$$(x-1)^{10} = \sum_{r=0}^{10} {}^{10}C_r x^{10-r} (-1)^r$$

$$x^7 \text{ term} = {}^8C_1 x^{8-1} (-1)^1 - {}^{10}C_3 x^{10-3} (-1)^3$$

$$\begin{aligned} \text{Coefficient of } x^7 &= -{}^8C_1 + {}^{10}C_3 \\ &= 112 \end{aligned}$$

Answer: C

- 8 What is the unit vector in the direction of $\overrightarrow{AB}$, where A and B are the points $(-2, 1)$ and $(6, 3)$, respectively?

A. $\frac{4}{\sqrt{17}}\mathbf{i} + \frac{1}{\sqrt{17}}\mathbf{j}$

B. $8\mathbf{i} + 2\mathbf{j}$

C. $\frac{2}{17}\mathbf{i} + \frac{1}{34}\mathbf{j}$

D. $\frac{2}{\sqrt{5}}\mathbf{i} + \frac{1}{\sqrt{5}}\mathbf{j}$

$$\begin{aligned}\overrightarrow{AB} &= 6\mathbf{i} + 3\mathbf{j} - (-2\mathbf{i} + \mathbf{j}) \\ &= 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{i} - \mathbf{j} \\ &= 8\mathbf{i} + 2\mathbf{j}\end{aligned}$$

$$\begin{aligned}|\overrightarrow{AB}| &= \sqrt{8^2 + 2^2} \\ &= 2\sqrt{17}\end{aligned}$$

$$\begin{aligned}\text{Unit vector} &= \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} \\ &= \frac{8\mathbf{i} + 2\mathbf{j}}{2\sqrt{17}} \\ &= \frac{4}{\sqrt{17}}\mathbf{i} + \frac{1}{\sqrt{17}}\mathbf{j}\end{aligned}$$

Answer: A

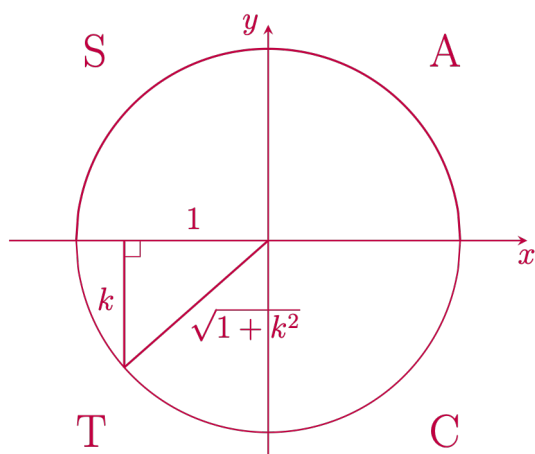
9 Given that $\tan x = k$, where $\pi < x < \frac{3\pi}{2}$, determine the value of $\sin \frac{x}{2}$.

A. $-\sqrt{\frac{1}{2}\left(1 + \frac{k}{\sqrt{1+k^2}}\right)}$

B. $\sqrt{\frac{1}{2}\left(1 + \frac{k}{\sqrt{1+k^2}}\right)}$

C. $-\sqrt{\frac{1}{2}\left(1 + \frac{1}{\sqrt{1+k^2}}\right)}$

D. $\sqrt{\frac{1}{2}\left(1 + \frac{1}{\sqrt{1+k^2}}\right)}$



$$\cos x = 1 - 2 \sin^2 \frac{x}{2}$$

$$-\frac{1}{\sqrt{1+k^2}} = 1 - 2 \sin^2 \frac{x}{2}$$

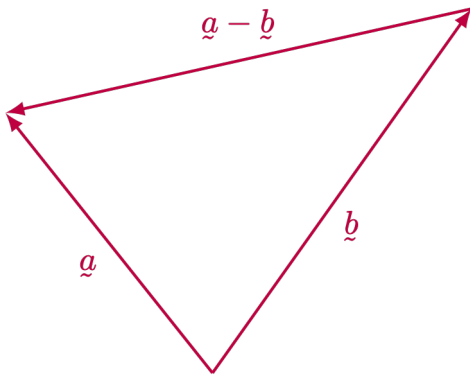
$$\sin^2 \frac{x}{2} = \frac{1}{2} \left(1 + \frac{1}{\sqrt{1+k^2}} \right)$$

$$\sin \frac{x}{2} = \sqrt{\frac{1}{2} \left(1 + \frac{1}{\sqrt{1+k^2}} \right)} \quad \left(\text{since } \sin \frac{x}{2} > 0, \text{ where } \frac{\pi}{2} < \frac{x}{2} < \frac{3\pi}{4} \right)$$

Answer: D

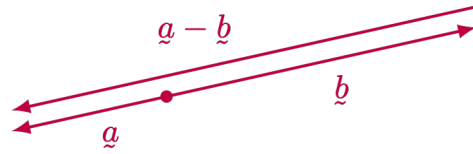
10 If $|\underline{a} - \underline{b}| = |\underline{a}| + |\underline{b}|$, where $\underline{a} \neq \underline{0}$ and $\underline{b} \neq \underline{0}$, which one of the following must be true?

- A. $|\underline{a}| = |\underline{b}|$
- B. $\underline{a}$ is parallel to $\underline{b}$
- C. $\underline{a} = \underline{b}$
- D. $\underline{a} \perp \underline{b}$



$\Rightarrow$

When $|\underline{a} - \underline{b}| = |\underline{a}| + |\underline{b}|$



$\therefore \underline{a} \parallel \underline{b}$

Answer: B

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section.

- Answer the questions in the spaces provided.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a separate Writing Booklet

(a) Solve $|4x - 3| > 11$.

$$\begin{array}{ll} 4x - 3 > 11 & \text{or} \quad 4x - 3 < -11 \\ 4x > 14 & \text{or} \quad 4x < -8 \\ x > \frac{7}{2} & \text{or} \quad x < -2 \end{array}$$

①

①

1 mark for each correct solution.

Question was done well.

(b) A cup of coffee is heated to 90°C . The room temperature is maintained at a constant 20°C , and the coffee cools to 50°C after 8 minutes. The temperature of the coffee can be modelled by the equation $T = 20 + 70e^{-kt}$, where T is the temperature of the coffee in degrees Celsius, t is the time in minutes since the coffee was placed in the room, and k is a constant.

Determine the time taken for the temperature of the coffee to reach 25°C .

Give your answer correct to 1 decimal place.

$$T = 20 + 70e^{-kt}$$

Sub. $t = 8$ and $T = 50$,

$$50 = 20 + 70e^{-8k}$$

$$70e^{-8k} = 30$$

$$e^{-8k} = \frac{3}{7}$$

$$-8k = \ln \frac{3}{7}$$

$$\therefore k = -\frac{1}{8} \ln \frac{3}{7}$$

$$\therefore T = 20 + 70e^{\frac{1}{8}t \ln \frac{3}{7}}$$

When $T = 25$,

$$25 = 20 + 70e^{\frac{1}{8}t \ln \frac{3}{7}}$$

$$e^{\frac{1}{8}t \ln \frac{3}{7}} = \frac{1}{14}$$

$$\frac{1}{8}t \ln \frac{3}{7} = \ln \frac{1}{14}$$

$$t = 8 \ln \frac{1}{14} \div \ln \frac{3}{7}$$

$$t = 24.9 \text{ minutes (1 d.p.)}$$

Question was done well.

many students wrote '14' then wrote '4' in the next line.

- (c) A solid is formed by rotating the region bounded by the curves $y = 4 - x^2$ and $y = 2 - x$ about the x -axis. Find the exact value of the volume of the solid.

4

① calculate intersection points

$$\begin{aligned} \text{For the points of intersection,} \\ 4 - x^2 &= 2 - x \\ x^2 - x - 2 &= 0 \\ (x - 2)(x + 1) &= 0 \\ x &= 2, -1 \end{aligned}$$

common mistake: students calculated intersection points for x correctly and then used '0 to 2' as limits instead.

① sub correct form and simplify

$$\begin{aligned} V &= \pi \int_{-1}^2 (4 - x^2)^2 dx - \pi \int_{-1}^2 (2 - x)^2 dx \\ &= \pi \int_{-1}^2 ((4 - x^2)^2 - (2 - x)^2) dx \\ &= \pi \int_{-1}^2 (16 - 8x^2 + x^4 - (4 - 4x + x^2)) dx \\ &= \pi \int_{-1}^2 (x^4 - 9x^2 + 4x + 12) dx \end{aligned}$$

common mistake: students didn't collect terms properly.

① integration step

$$\begin{aligned} &= \pi \left[\frac{x^5}{5} - 3x^3 + 2x^2 + 12x \right]_{-1}^2 \\ &= \pi \left[\left(\frac{32}{5} - 24 + 8 + 24 \right) - \left(-\frac{1}{5} + 3 + 2 - 12 \right) \right] \\ &= \frac{108\pi}{5} \text{ unit}^3 \end{aligned}$$

① final answer

- (d) (i) Complete the square for $x^2 - 6x + 13$, expressing the answer in the form $[f(x)]^2 + a$, where a is a constant and $f(x)$ is a linear function of x .

1

$$\begin{aligned} x^2 - 6x + 13 &= x^2 - 6x + 9 + 4 \\ &= (x - 3)^2 + 4 \end{aligned}$$

① completing the square.

(ii) Hence, find $\int \frac{1}{x^2 - 6x + 13} dx$.

2

$$\begin{aligned} \int \frac{1}{x^2 - 6x + 13} dx &= \int \frac{1}{4 + (x - 3)^2} dx \\ &= \frac{1}{2} \tan^{-1} \frac{x - 3}{2} + C \end{aligned}$$

(from (a)) ① correct sub
① using correct identity.

Question was done well

(e) A standard pack of 52 playing cards consists of 13 hearts, 13 diamonds, 13 clubs and 13 spades. Fifteen cards are drawn with replacement from the pack.

- (i) Let X be the number of diamonds drawn. Find the mean and standard deviation of X , giving your answers correct to 2 decimal places, where applicable. 2

$$\begin{aligned} n &= 15, & p &= \frac{13}{52}, & q &= 1 - \frac{1}{4} \\ & & &= 0.25, & &= 0.75 \\ \mu &= np, & \sigma^2 &= npq \\ &= 15 \times 0.25 & &= 15 \times 0.25 \times 0.75 \\ &= 3.75 & \sigma &= \sqrt{2.8125} \\ & & &= 1.68 \end{aligned}$$

Question was done well.

Common mistake: students did not square root variance.

1 mark each, mean and standard deviation

- (ii) Determine the probability that X is within one standard deviation of the mean. Give your answer correct to 3 decimal places. 2

$$\begin{aligned} P(3.75 - 1.68 \leq X \leq 3.75 + 1.68) \\ &= P(2.07 \leq X \leq 5.43) \\ &= P(X = 3) + P(X = 4) + P(X = 5) \\ &= {}^{15}C_3(0.25)^3(0.75)^{12} + {}^{15}C_4(0.25)^4(0.75)^{11} + {}^{15}C_5(0.25)^5(0.75)^{10} \\ &= 0.616 \end{aligned}$$

1

± standard deviation from mean.

1 correctly calculate probability.

Overall done well

• Some students only calculated top limit. (no marks)

• Many students calculated $P(2.07 \leq X \leq 5.43)$ and didn't continue
or calculated $P(X=2) + P(X=3) + P(X=4) + P(X=5) + P(X=6)$

• Rounding should have been better.

Question 12 (15 marks) Use a separate Writing Booklet

- (a) There are 5 men and 6 women to choose from. In how many ways can 3 men and 5 women be selected and arranged in a row? 2

Done very well ✓

A group of 3 men and 5 women can be chosen in ${}^5C_3 \times {}^6C_5$ ways.

Since each group can be arranged in a row in $8!$ ways,

$$\text{Number of arrangements} = \frac{{}^5C_3 \times {}^6C_5 \times 8!}{1} = 2419200 \quad (1)$$

- (b) Consider the function $f(x) = \sqrt{3} \cos x + 3 \sin x$ for $0 \leq x \leq 2\pi$.

- (i) By first expanding $\cos(x - \alpha)$, express $f(x)$ in the form $R \cos(x - \alpha)$. 3

$$\begin{aligned} \sqrt{3} \cos x + 3 \sin x &= R \cos(x - \alpha) \\ &= R \cos x \cos \alpha + R \sin x \sin \alpha \end{aligned}$$

Equate coefficients,

$$R \cos \alpha = \sqrt{3} \quad [1]$$

$$R \sin \alpha = 3 \quad [2]$$

$$[1]^2 + [2]^2, \quad R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 12$$

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 12$$

$$R^2 = 12$$

$$\therefore R = 2\sqrt{3} \quad (\text{since } R > 0) \quad (1)$$

$$\text{From [1], } \cos \alpha = \frac{\sqrt{3}}{2\sqrt{3}} = \frac{1}{2} > 0$$

$$\text{From [2], } \sin \alpha = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2} > 0$$

* Since $\cos \alpha > 0$ and $\sin \alpha > 0$, α is in the 1st quadrant with related angle $\frac{\pi}{3}$.

$$\therefore \alpha = \frac{\pi}{3} \quad (1)$$

$$\therefore \sqrt{3} \cos x + 3 \sin x = 2\sqrt{3} \cos \left(x - \frac{\pi}{3}\right) \quad (1)$$

Many students did not justify why angle was in the first quadrant (*)
→ Did not penalise.

- (ii) Hence, solve $\sqrt{3} \cos x + 3 \sin x = 1$, correct to 2 decimal places. 2

$$2\sqrt{3} \cos \left(x - \frac{\pi}{3}\right) = 1, \text{ where } -\frac{\pi}{3} \leq x - \frac{\pi}{3} \leq \frac{5\pi}{3}$$

$$\cos \left(x - \frac{\pi}{3}\right) = \frac{1}{2\sqrt{3}}$$

$$x - \frac{\pi}{3} = \cos^{-1} \frac{1}{2\sqrt{3}} \quad (1)$$

$$= 1.278$$

Since $\cos \left(x - \frac{\pi}{3}\right) > 0$, $x - \frac{\pi}{3}$ lies in the first and fourth quadrants.

$$x - \frac{\pi}{3} = 1.278 \text{ or } 2\pi - 1.278$$

$$\therefore x = 2.33 \text{ or } 6.05 \quad (1)$$

Done very well ✓

(c) Express $\sin\left(2\cos^{-1}\frac{3}{4}\right)$ in the form $a\sqrt{b}$, where a and b are rational numbers.

2

Let $\theta = \cos^{-1}\frac{3}{4}$
Then $\cos\theta = \frac{3}{4}$



①

$$\begin{aligned}\text{Opposite} &= \sqrt{4^2 - 3^2} \\ &= \sqrt{7}\end{aligned}$$

$$\sin\theta = \frac{\sqrt{7}}{4} \quad (\text{since } \cos\theta > 0 \text{ and } 0 \leq \cos^{-1}x \leq \pi)$$

$$\begin{aligned}\sin\left(2\cos^{-1}\frac{3}{4}\right) &= \sin(2\theta) \\ &= 2\sin\theta\cos\theta \\ &= 2\left(\frac{\sqrt{7}}{4}\right)\left(\frac{3}{4}\right) \\ &= \frac{3}{8}\sqrt{7} \quad \text{①}\end{aligned}$$

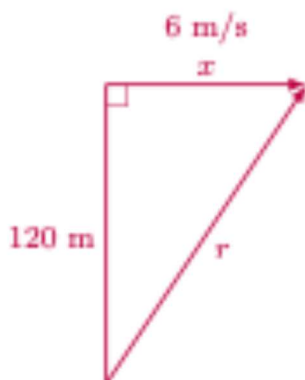
Done mostly well.
→ Some didn't draw the correct triangle which messed up the whole solution.

(d) A drone is flying horizontally at a constant height of 120 m above the ground.

3

It moves horizontally away from a person at a constant speed of 6 m/s.

Determine the rate at which the distance between the drone and the person is increasing when the drone is 150 m from the person.



$$\frac{dx}{dt} = 6, \quad \frac{dr}{dt} = ?, \quad r^2 = x^2 + 120^2$$

$$\begin{aligned}r &= \sqrt{x^2 + 120^2} \quad \text{①} \\ \frac{dr}{dx} &= \frac{1}{2}(x^2 + 14400)^{-\frac{1}{2}}(2x) \\ &= \frac{x}{\sqrt{x^2 + 14400}}\end{aligned}$$

$$\begin{aligned}\frac{dr}{dt} &= \frac{dr}{dx} \times \frac{dx}{dt} \\ &= \frac{x}{\sqrt{x^2 + 14400}} \times 6 \quad \text{①} \\ &= \frac{6x}{\sqrt{x^2 + 14400}}\end{aligned}$$

• Most students reviewed 2/3 marks.

• Common error : - $x = 150$.

- Incorrect related rate (reciprocal).

$$\begin{aligned}\text{When } r &= 150, \quad 150^2 = x^2 + 120^2 \\ x^2 &= 8100 \\ x &= 90\end{aligned}$$

$$\begin{aligned}\frac{dr}{dt} &= \frac{6 \times 90}{\sqrt{90^2 + 14400}} \\ &= \frac{540}{\sqrt{22500}} \\ &= \frac{18}{5} \text{ m/s} \quad \text{①}\end{aligned}$$

(e) Using the substitution $u = \sqrt{6x - 1}$, show that

3

$$\int \frac{x}{\sqrt{6x-1}} dx = \frac{1}{27}(3x+1)\sqrt{6x-1} + C.$$

$$\begin{aligned} \int \frac{x}{\sqrt{6x-1}} dx &= \frac{1}{18} \int (u^2 + 1) du \\ &= \frac{1}{18} \left(\frac{u^3}{3} + u \right) + C \quad \textcircled{1} \\ &= \frac{1}{18} \left(\frac{(\sqrt{6x-1})^3}{3} + \sqrt{6x-1} \right) + C \\ &= \frac{1}{54} \left((6x-1)\sqrt{6x-1} + 3\sqrt{6x-1} \right) + C \\ &= \frac{1}{54} \sqrt{6x-1} (6x-1+3) + C \\ &= \frac{1}{54} \sqrt{6x-1} (6x+2) + C \\ &= \frac{1}{27} (3x+1)\sqrt{6x-1} + C \quad \textcircled{1} \end{aligned}$$

$u = \sqrt{6x-1}, \quad u^2 = 6x-1, \quad x = \frac{1}{6}(u^2+1)$

$\frac{du}{dx} = \frac{1}{2}(6x-1)^{-\frac{1}{2}} \times 6$

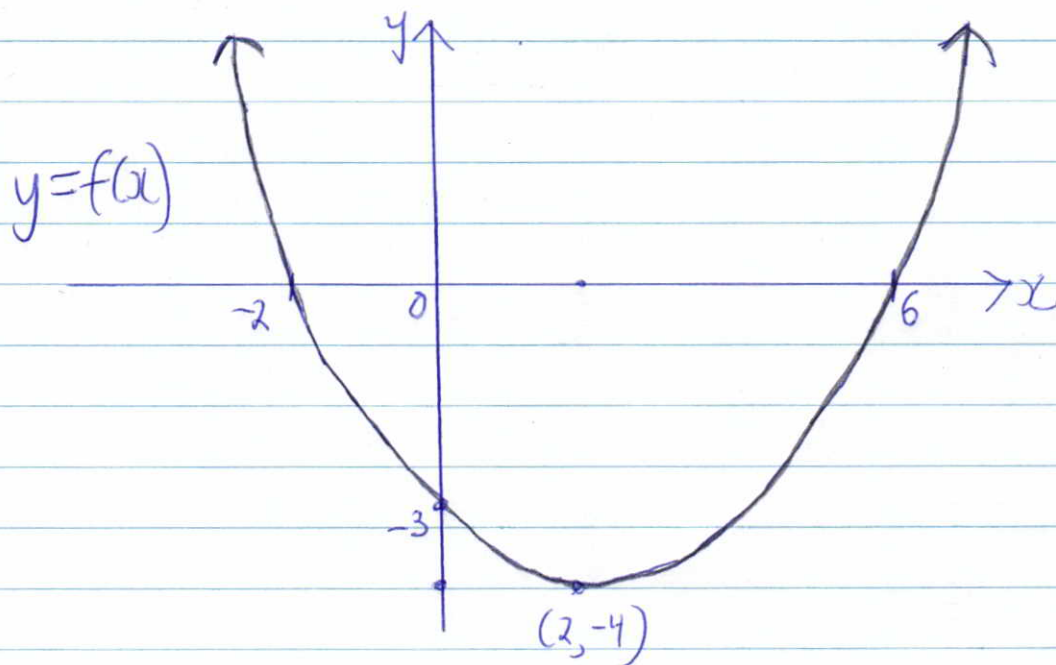
$\frac{du}{dx} = \frac{3}{\sqrt{6x-1}}$

$\frac{1}{3} du = \frac{1}{\sqrt{6x-1}} dx$

• Done well ✓

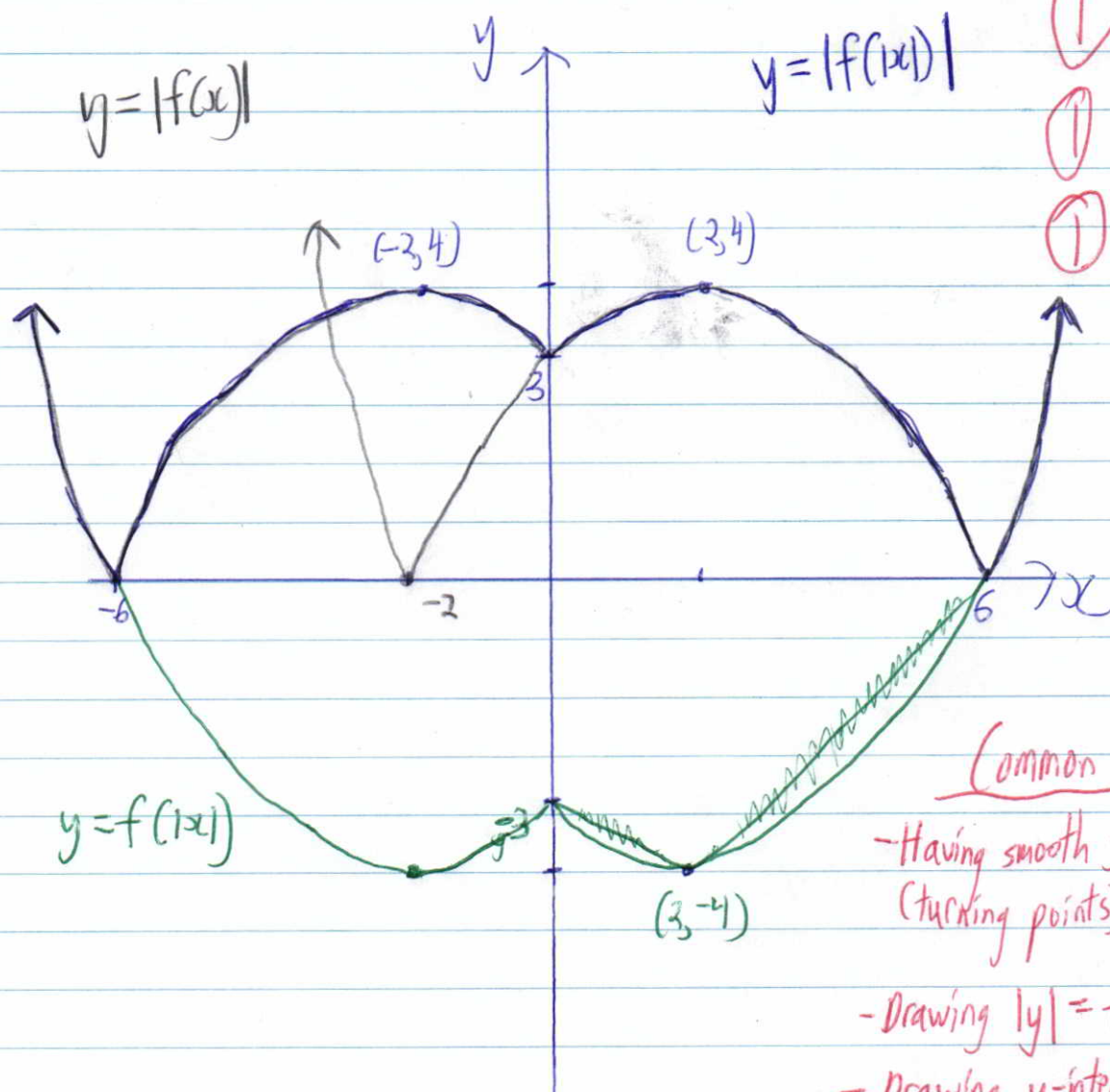
• Do not skip too many steps!

Q3a



Criteria

- ① $|f(x)|$
- ① $f(|x|)$
- ① key points/shape



Common Errors:

- Having smooth x-intercepts (turning points), not sharp changes
- Drawing $|y| = f(|x|)$
- Drawing y-intercept at the origin not $y = 3$
- Poor scales (2 is not half of 6)

Q13b

Step 1: Prove true for $n=1$

$$\begin{aligned}4(3^1) + 8^1 &= 12 + 8 \\&= 20 \\&= 5(4) \quad (\text{Divisible by } 5)\end{aligned}$$

① Base case

$\therefore$ the statement is true for $n=1$

Step 2: Assume true for $n=k$ where k is an integer ≥ 1

$$\begin{aligned}\text{i.e. assume } 4(3^k) + 8^k &= 5P, P \in \mathbb{Z} \\8^k &= 5P - 4(3^k)\end{aligned}$$

Alternate:

① $4(3^k) = 5P - 8^k$
Assumption and RTP

Step 3: Given assumption, prove true for $n=k+1$

$$\text{i.e. prove } 4(3^{k+1}) + 8^{k+1} = 5Q \quad (Q \in \mathbb{Z})$$

$$\begin{aligned}\text{LHS} &= 4(3^{k+1}) + 8^{k+1} = 4(3^k \times 3) + 8(8^k) \\&= 12(3^k) + 8(5P - 4(3^k)) \quad (\text{Using assumption}) \\&= 12(3^k) + 40P - 32(3^k) \\&= 40P - 20(3^k) \\&= 5(8P - 4(3^k)) \\&= 5Q \quad (Q = 8P - 4(3^k) \in \mathbb{Z})\end{aligned}$$

① Using assumption

(Using assumption)

(Using assumption)

$$\begin{aligned}&= 3(5P - 8^k) + 8(8^k) \\&= 15P - 3(8^k) + 8(8^k) \\&= 15P + 5(8^k) \\&= 5(3P + 8^k) \\&= 5Q \quad (Q = 3P + 8^k \in \mathbb{Z})\end{aligned}$$

$\therefore$ if the result is true for $n=k$, then it is also true for $n=k+1$.

① Simplifies and conclusion

Step 4: It follows from steps 1, 2, and 3, by the principle of mathematical induction, the statement is true for all integers $n \geq 1$

$$\begin{aligned}4(3^1) + 8^1 &\neq 15 \\4(3^k \times 3) &\neq 12 \times 12^k\end{aligned}$$

Feedback

- Done well
- most mistakes were silly substitution, factorisation, or algebra
- too many students still not writing (using assumption)

Q13c

$$= 40P - 20(3^k)$$

$$= 5(8P - 4(3^k)), \text{ which is divisible by 5.}$$

Step 3 By the principle of mathematical induction, the statement is true for all integers $n \geq 0$.

Feedback

(c) Consider the differential equation $\frac{dy}{dx} = \frac{y(1-x)}{x}$.

(i) State any constant solutions.

Substitute $\frac{dy}{dx} = 0$ for the constant solution,

$$\frac{y(1-x)}{x} = 0$$

$$\therefore y = 0$$

- Poorly done for such an easy question. 1

- Constant solutions are constant functions (horizontal lines)

- No mark if $x=0$ was included

(ii) Find the general solution of the differential equation by separating the variables. 3

$$\frac{dy}{dx} = \frac{y(1-x)}{x}$$

$$\frac{1}{y} dy = \frac{1-x}{x} dx \quad - \textcircled{1} \text{ Separation of variables}$$

Integrating both sides,

$$\int \frac{1}{y} dy = \int \frac{1-x}{x} dx$$

$$\int \frac{1}{y} dy = \int \left(\frac{1}{x} - 1 \right) dx$$

$$\ln|y| = \ln|x| - x + C \quad - \textcircled{1} \text{ Integrates}$$

$$\ln|y| - \ln|x| = -x + C$$

$$\ln \left| \frac{y}{x} \right| = -x + C$$

$$\left| \frac{y}{x} \right| = e^{-x+C}$$

$$\frac{y}{x} = \pm e^C e^{-x}$$

$$\therefore y = A x e^{-x}, \quad \text{where } A \text{ is a nonzero constant}$$

- \textcircled{1} Makes y the subject

not correct if you remove either $|y|$ or $|x|$ without explanation

Feedback:

- Poor mathematical understanding overall, despite decent attempts
- Integrate both sides at the same time. Otherwise you should have 2 constants (not penalised)
- $\pm A x e^{-x}$ ($A = e^C > 0$) is not a proper set of general solutions. Each constant should give 1 curve, not 2 (not penalised)
- Many students still don't properly understand how to deal with absolute values and arbitrary constants

Q3c - Alternate

$$\text{ii) } \frac{dy}{dx} = \frac{y(1-x)}{x}$$

$$\frac{1}{y} dy = \left(\frac{1}{x} - 1\right) dx \quad (\text{Variable separable}) \quad (1)$$

$$\int \frac{1}{y} dy = \int \left(\frac{1}{x} - 1\right) dx$$

$$\log_e |y| = \log_e |x| - x + C \quad (1)$$

$$|y| = e^{\log_e |x| - x + C}$$

$$|y| = e^{\log_e |x|} \cdot e^{-x} \cdot e^C$$

$$|y| = A |x| e^{-x} \quad (A = e^C > 0)$$

$$\therefore y = A |x| e^{-x} \quad (A \text{ is a positive or negative constant}) \quad (1)$$

$|y| = e^C |x| e^{-x}$ to $y = A x e^{-x}$ is not valid working

- If you did not combine absolute values, the $|x|$ cannot ~~disappear~~ disappear without explanation

$$|y| = |x| f(x) \rightarrow \left|\frac{y}{x}\right| = f(x)$$

$$\frac{y}{x} = \pm f(x)$$

$$y = \pm x f(x)$$

Q13d

- (d) Find all possible values of k given that the length of the projection of $\mathbf{u}$ onto $\mathbf{v}$ is $\frac{108}{17}$,
where $\mathbf{u} = k\mathbf{i} + 9\mathbf{j}$ and $\mathbf{v} = 15\mathbf{i} - 8\mathbf{j}$.

4

$$\begin{aligned}\text{proj}_{\mathbf{v}} \mathbf{u} &= \frac{(\mathbf{u} \cdot \mathbf{v})\mathbf{v}}{|\mathbf{v}||\mathbf{v}|} \\ &= \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \frac{\mathbf{v}}{|\mathbf{v}|} \\ &= \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \hat{\mathbf{v}}\end{aligned}$$

Length of the projection = $\left| \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \right|$ - ① Uses length of projection

$$\left| \frac{(k\mathbf{i} + 9\mathbf{j}) \cdot (15\mathbf{i} - 8\mathbf{j})}{\sqrt{15^2 + (-8)^2}} \right| = \frac{108}{17}$$

$$\left| \frac{15k - 72}{17} \right| = \frac{108}{17}$$

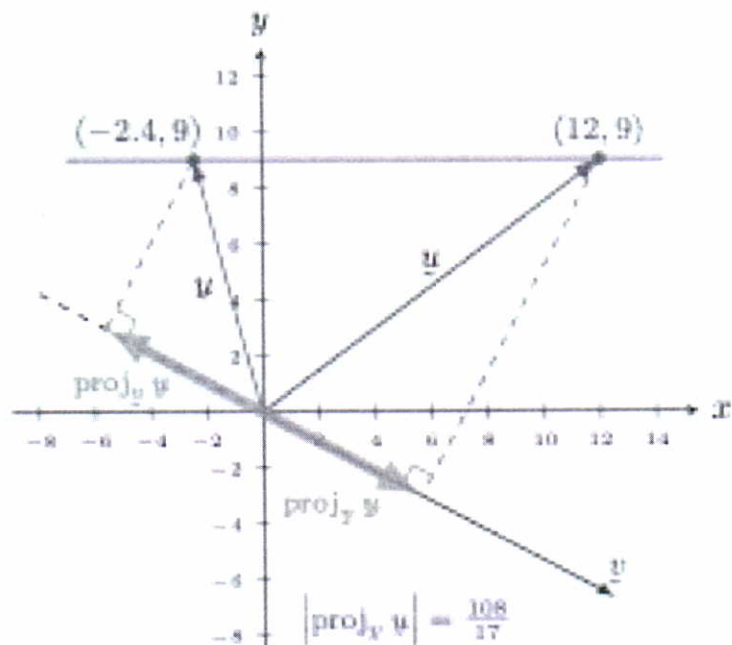
- ① simplifies

$$|15k - 72| = 108$$

$$15k - 72 = 108 \quad \text{or} \quad 15k - 72 = -108$$

$$15k = 180 \quad \text{or} \quad 15k = -36$$

$$\therefore k = 12 \text{ - ① or } k = -\frac{12}{5} \text{ (or } -2.4) \text{ - ①}$$



Feedback:

- Many students forget absolute values.
- Length means absolute value and $15k - 72$ is not known to be positive.

Q13d - Alternate

$$\text{proj}_{\underline{v}} \underline{u} = \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \underline{v}$$

$$= \frac{(k\underline{i} + 9\underline{j}) \cdot (15\underline{i} - 8\underline{j})}{(\sqrt{15^2 + (-8)^2})^2} (15\underline{i} - 8\underline{j})$$

$$= \frac{15k - 72}{17^2} (15\underline{i} - 8\underline{j})$$

$$\text{Let } |\text{proj}_{\underline{v}} \underline{u}| = \frac{108}{17}$$

$$\left| \frac{15k - 72}{17^2} (15\underline{i} - 8\underline{j}) \right| = \frac{108}{17} \rightarrow \text{If fully expanded: Equation: } 65025k^2 - 624240k - 1872720 = 0$$

$$\left| \frac{1}{17^2} (225k - 1080) \right| = \frac{108}{17}$$

① Writes equation for projection

$$\left| \begin{pmatrix} 225k - 1080 \\ -120k + 576 \end{pmatrix} \right| = 17 \times 108$$

$$\begin{aligned} 5k^2 - 48k - 144 &= 0 \\ k &= \frac{48 \pm \sqrt{(-48)^2 - 4(5)(-144)}}{2(5)} \\ &= \frac{48 \pm 72}{10} \\ &= 12 \text{ or } -\frac{12}{5} \end{aligned}$$

$$\sqrt{(225k - 1080)^2 + (-120k + 576)^2} = 17 \times 108$$

$$(225k - 1080)^2 + (-120k + 576)^2 = 17^2 \times 108^2$$

$$15^2(15k - 72)^2 + 8^2(-15k + 72)^2 = 17^2 \times 108^2$$

$$17^2(15k - 72)^2 = 108^2 \times 17^2$$

$$15k - 72 = \pm 108$$

① Simplifies equation OR

$$\therefore 15k - 72 = 108 \quad \text{or} \quad 15k - 72 = -108$$

$$15k = 180$$

$$k = 12$$

$$15k = -36$$

$$k = -\frac{12}{5}$$

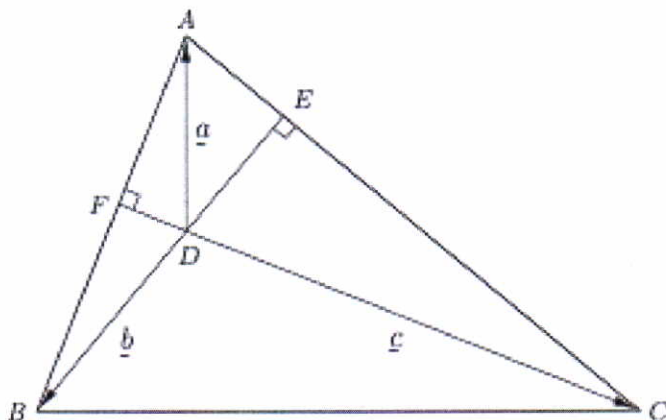
① Finds solutions

The majority of students who used the full formula, particularly the many that did the green method made algebra/calculation errors.

The formula: $\text{proj}_{\underline{v}} \underline{u} = \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|} \hat{\underline{v}}$ (multiple of a unit vector) should be known

Question 14 (15 marks) Use a separate Writing Booklet

- (a) The $\triangle ABC$ shown below has altitudes BE and CF , where an altitude is a line segment drawn from a vertex perpendicular to the opposite side. The altitudes intersect at D .
Let $\overrightarrow{DA} = \underline{a}$, $\overrightarrow{DB} = \underline{b}$ and $\overrightarrow{DC} = \underline{c}$.



- (i) Show that $\underline{a} \cdot \underline{b} = \underline{a} \cdot \underline{c} = \underline{b} \cdot \underline{c}$.

$$\overrightarrow{AC} \cdot \overrightarrow{DB} = 0 \quad (\text{since } AC \perp BE)$$

$$(\underline{c} - \underline{a}) \cdot \underline{b} = 0$$

$$\underline{c} \cdot \underline{b} - \underline{a} \cdot \underline{b} = 0$$

$$\therefore \underline{b} \cdot \underline{c} = \underline{a} \cdot \underline{b}$$

$$\overrightarrow{AB} \cdot \overrightarrow{DC} = 0 \quad (\text{since } AB \perp FC)$$

$$(\underline{b} - \underline{a}) \cdot \underline{c} = 0$$

$$\underline{b} \cdot \underline{c} - \underline{a} \cdot \underline{c} = 0$$

$$\therefore \underline{b} \cdot \underline{c} = \underline{a} \cdot \underline{c}$$

$$\therefore \underline{a} \cdot \underline{b} = \underline{a} \cdot \underline{c} = \underline{b} \cdot \underline{c}$$

3
Many students started with $\overrightarrow{AC} \perp \overrightarrow{BE}$, then label $\overrightarrow{DB} = \lambda \overrightarrow{BE}$ and made it complicated.
- overall, it was not done well.

- (ii) Hence, or otherwise, prove that the three altitudes of $\triangle ABC$ are concurrent; that is, they all pass through a single point.

$$\overrightarrow{BC} \cdot \overrightarrow{DA} = (\underline{c} - \underline{b}) \cdot \underline{a}$$

$$= \underline{c} \cdot \underline{a} - \underline{b} \cdot \underline{a}$$

$$= \underline{c} \cdot \underline{a} - \underline{c} \cdot \underline{a} \quad (\text{since } \underline{a} \cdot \underline{b} = \underline{a} \cdot \underline{c})$$

$$= 0$$

$$\therefore BC \perp AD$$

$\therefore$ The extension of AD to meet BC is an altitude of the $\triangle ABC$.

$\therefore$ The three altitudes of $\triangle ABC$ are concurrent.

2
similarly, many students introduced λ, μ for $\overrightarrow{BC} \perp \overrightarrow{DA}$, and could not arrive at the final step.

(b) A ball is projected from the base of a cliff with initial speed u m/s at an angle of θ to the horizontal. At the same time, a second ball is projected horizontally with initial speed w m/s from the top of the same cliff, which is h m high. Take the base of the cliff as the origin $(0,0)$. The two balls collide.

(i) Show that the position vector of the first ball is

3

$$\mathbf{r}_1(t) = (ut \cos \theta) \mathbf{i} + \left(ut \sin \theta - \frac{g}{2} t^2 \right) \mathbf{j}.$$

$$\mathbf{a}_1(t) = -g \mathbf{j}$$

$$\mathbf{v}_1(t) = \int (-g \mathbf{j}) dt$$

$$= -gt \mathbf{j} + \mathbf{c}_1$$

$$\text{Sub. } t = 0 \text{ and } \mathbf{v}_1(0) = u \cos \theta \mathbf{i} + u \sin \theta \mathbf{j},$$

$$\therefore \mathbf{c}_1 = u \cos \theta \mathbf{i} + u \sin \theta \mathbf{j}$$

$$\therefore \mathbf{v}_1 = -gt \mathbf{j} + u \cos \theta \mathbf{i} + u \sin \theta \mathbf{j}$$

$$= u \cos \theta \mathbf{i} + (u \sin \theta - gt) \mathbf{j}$$

$$\mathbf{r}_1 = \int (u \cos \theta \mathbf{i} + (u \sin \theta - gt) \mathbf{j}) dt$$

$$= ut \cos \theta \mathbf{i} + \left(ut \sin \theta - \frac{1}{2} gt^2 \right) \mathbf{j} + \mathbf{d}_1$$

$$\text{Sub. } t = 0 \text{ and } \mathbf{r}_1(0) = \mathbf{0},$$

$$\therefore \mathbf{d}_1 = \mathbf{0}$$

$$\therefore \mathbf{r}_1 = ut \cos \theta \mathbf{i} + \left(ut \sin \theta - \frac{g}{2} t^2 \right) \mathbf{j}$$

comments :

this part was done well, except

- few students forgot to find constants for velocity and displacement.

- overall okay.

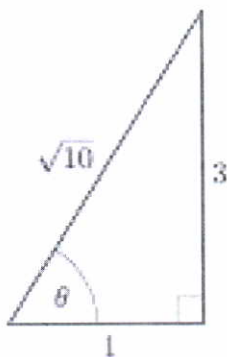
(ii) It is given that $\theta = \tan^{-1} 3$ and the position vector of the second ball is

2

$$\mathbf{r}_2(t) = w t \mathbf{i} + \left(h - \frac{g}{2} t^2 \right) \mathbf{j}.$$

Show that $\cos \theta = \frac{1}{\sqrt{10}}$, and hence deduce that $u = \sqrt{10}w$.

$$\tan \theta = 3$$



$$\text{hypotenuse} = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$\therefore \sin \theta = \frac{3}{\sqrt{10}}$$

$$\therefore \cos \theta = \frac{1}{\sqrt{10}}$$

- well done !!

At the point of collision, $\mathbf{r}_1(t) = \mathbf{r}_2(t)$.

Equating x components,

$$ut \cos \theta = wt$$

$$u \frac{1}{\sqrt{10}} = w$$

$$\therefore u = \sqrt{10}w$$

(iii) Hence, prove that the balls collide at time $t = \frac{h}{3w}$, and that the height of the balls at the point of collision is $h - \frac{gh^2}{18w^2}$.

2

From $\mathbf{r}_1(t) = \mathbf{r}_2(t)$, equating y components,

$$ut \sin \theta - \frac{g}{2}t^2 = h - \frac{g}{2}t^2$$

$$ut \sin \theta = h$$

$$\sqrt{10}w \left(\frac{3}{\sqrt{10}} \right) t = h$$

$$\therefore t = \frac{h}{3w}$$

well done !!

Substitute $t = \frac{h}{3w}$ into $y_2 = h - \frac{g}{2}t^2$ for the height,

$$y_2 = h - \frac{g}{2} \left(\frac{h}{3w} \right)^2$$

$$\therefore y_2 = h - \frac{gh^2}{18w^2}$$

well done !!

(iv) At the instant of collision, the velocity vectors of the two balls are perpendicular.

3

$$\text{Show that } h = \frac{3(3 \pm \sqrt{5})w^2}{2g}.$$

$$\begin{aligned} \text{At } t = \frac{h}{3w}, \quad \mathbf{v}_1(t) &= u \cos \theta \mathbf{i} + (u \sin \theta - gt) \mathbf{j} \\ &= \sqrt{10}w \left(\frac{1}{\sqrt{10}} \right) \mathbf{i} + \left(\sqrt{10}w \left(\frac{3}{\sqrt{10}} \right) - g \left(\frac{h}{3w} \right) \right) \mathbf{j} \\ &= w \mathbf{i} + \left(3w - \frac{gh}{3w} \right) \mathbf{j} \end{aligned}$$

$$\begin{aligned} \mathbf{v}_2(t) &= w \mathbf{i} - gt \mathbf{j} \\ &= w \mathbf{i} - g \left(\frac{h}{3w} \right) \mathbf{j} \\ &= w \mathbf{i} - \frac{gh}{3w} \mathbf{j} \end{aligned}$$

- some students forgot to write $\mathbf{v}_1(t) \cdot \mathbf{v}_2(t) = 0$

- some students arrive at the final step with their incorrect working.

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 0 \quad (\text{since } \mathbf{v}_1 \perp \mathbf{v}_2)$$

$$w^2 - \frac{gh}{3w} \left(3w - \frac{gh}{3w} \right) = 0$$

$$\left(\frac{gh}{3w} \right)^2 - 3w \left(\frac{gh}{3w} \right) + w^2 = 0$$

$$\frac{gh}{3w} = \frac{3w \pm \sqrt{9w^2 - 4w^2}}{2}$$

$$\frac{gh}{3w} = \frac{3w \pm \sqrt{5}w}{2}$$

$$\frac{gh}{3w} = \frac{(3 \pm \sqrt{5})w}{2}$$

$$\therefore h = \frac{3(3 \pm \sqrt{5})w^2}{2g}$$

- overall, it was not done well.

Alternative:

$$\frac{g^2}{9w^2}h^2 - gh + w^2 = 0$$

$$h = \frac{g \pm \sqrt{g^2 - \frac{4g^2}{9}}}{\frac{2g^2}{9w^2}}$$

$$= \left(g \pm g \sqrt{1 - \frac{4}{9}} \right) \times \frac{9w^2}{2g^2}$$

$$= \left(1 \pm \sqrt{\frac{5}{9}} \right) \times \frac{9w^2}{2g}$$

$$= \frac{9w^2 \left(1 \pm \frac{1}{3}\sqrt{5} \right)}{2g}$$

$$= \frac{3w^2(3 \pm \sqrt{5})}{2g}$$

$$\therefore h = \frac{3(3 \pm \sqrt{5})w^2}{2g}$$

- Few students used this method.

- Again ^{few} students show incorrect working to arrive at the final step.

- Overall, it was not done so well.

End of paper